Lab No: 4 Date:2025/6/4

Title: Understanding the basics of configuring a static IP address on Linux/Windows Machines.

Theory

1. **IP address:**

   An IP (Internet Protocol) address is a unique numerical label assigned to every device, such as a computer, tablet, or smartphone, participating in a computer network that uses the Internet Protocol for communication. It serves two main functions: host or network interface identification and location addressing.

   Example: A typical private IP address you might see at home is 192.168.1.10.

   i. **Static IP:**

      A static IP address is a permanent numerical label manually assigned to a device on a network. It remains constant and does not change over time unless it's manually reconfigured. Static IPs are often used for servers, printers, or other devices that need to be consistently reachable at the same address.

      Example: If you run a web server from our home, you might want a static IP address so users can always find our website at the same location.

   ii. **Dynamic IP:**

      A dynamic IP address is a temporary numerical label assigned to a device by a DHCP (Dynamic Host Configuration Protocol) server. These addresses are leased for a certain period and can change whenever the lease expires or the device reconnects to the network. Most home devices get dynamic IP addresses from their router.

      Example: When you connect laptop to Wi-Fi at a coffee shop, it's typically assigned a dynamic IP address, which might be different the next time you visit.

2. **Subnet mask:**

   A subnet mask is a 32-bit number used in conjunction with an IP address to divide a network into smaller, more manageable subnetworks. It tells a computer which part of an IP address refers to the network and which part refers to the host (individual device) within that network. This helps in routing data packets efficiently.

Example: For an IP address 192.168.1.10, a common subnet mask like 255.255.255.0 indicates that the "192.168.1" part identifies the network, and "10" identifies the specific device.

3. **Default Gateway:**

   The default gateway is a network device, often a router, that serves as a crucial point for data leaving the local network. When a device needs to send information to a destination outside its own local network (e.g., to the internet), it sends the data to the default gateway, which then forwards it to the correct destination.

   Example: Our home router's IP address, such as 192.168.1.1 or 192.168.1.254, is commonly set as the default gateway for devices on our network.

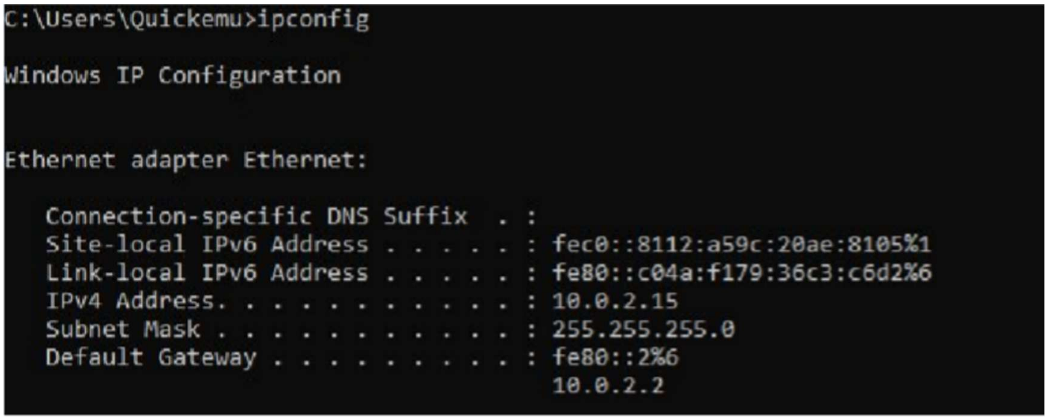
4. **DNS server:**

   A DNS server acts like a phonebook for the internet, translating human-friendly domain names (like "www.google.com") into machine-readable numerical IP addresses (like 142.250.190.174) that computers use to locate and communicate with each other. This process is essential for Browse the internet by domain names instead of remembering complex IP addresses.

   Example: When you type "www.youtube.com" into our browser, a DNS server translates it into an IP address so our computer can connect to YouTube's server. Well-known public DNS servers include Google DNS (8.8.8.8 and 8.8.4.4) and Cloudflare DNS (1.1.1.1).

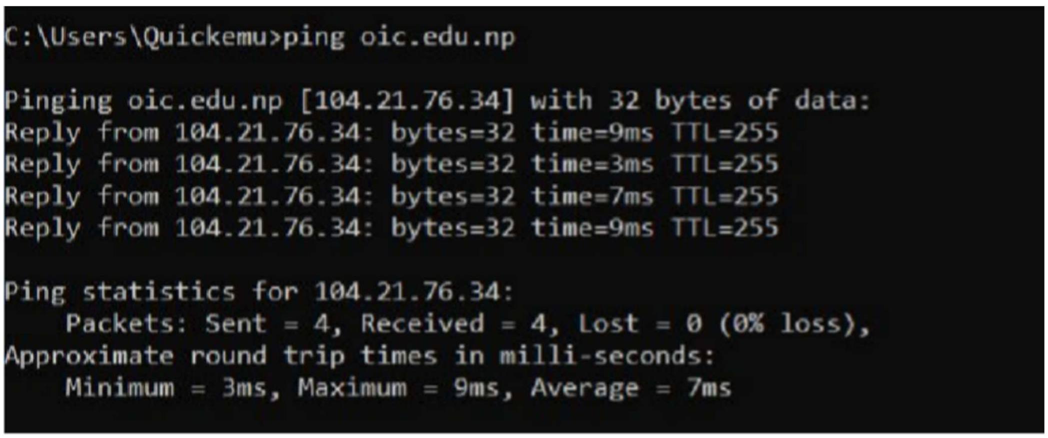
Static IP Configuration using GUI in Windows:

1. Test current device IP and connectivity:
   i. First test the device’s IP configuration using *ipconfig* command



```
C:\Users\Quickemu>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Site-local IPv6 Address . . . . . : fec0::8112:a59c:20ae:8105%1
   Link-local IPv6 Address . . . . . : fe80::c04a:f179:36c3:c6d2%6
   IPv4 Address. . . . . . . . . . . : 10.0.2.15
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2%6
                                       10.0.2.2
```

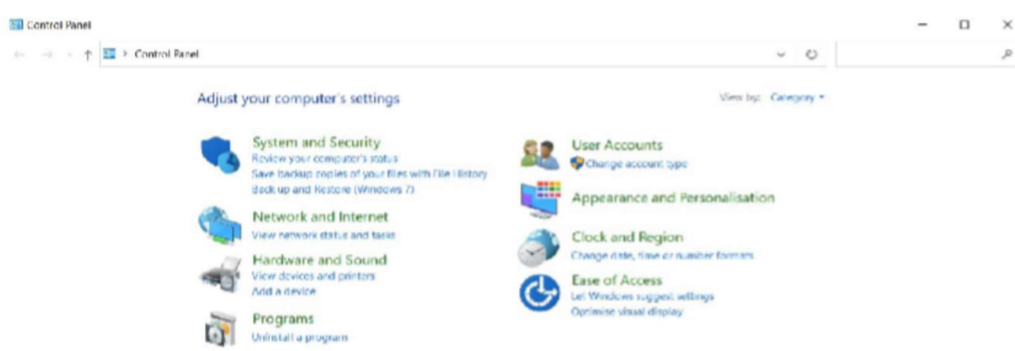
ii. Now, test the device's connectivity using the *ping* command.



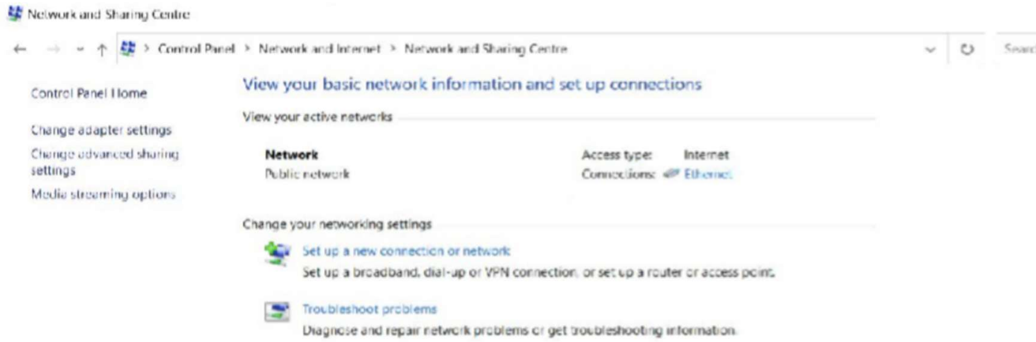
```
C:\Users\Quickemu>ping oic.edu.np

Pinging oic.edu.np [104.21.76.34] with 32 bytes of data:
Reply from 104.21.76.34: bytes=32 time=9ms TTL=255
Reply from 104.21.76.34: bytes=32 time=3ms TTL=255
Reply from 104.21.76.34: bytes=32 time=7ms TTL=255
Reply from 104.21.76.34: bytes=32 time=9ms TTL=255

Ping statistics for 104.21.76.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 9ms, Average = 7ms
```

2. Access Network adapter settings:

i. Open the Control Panel.



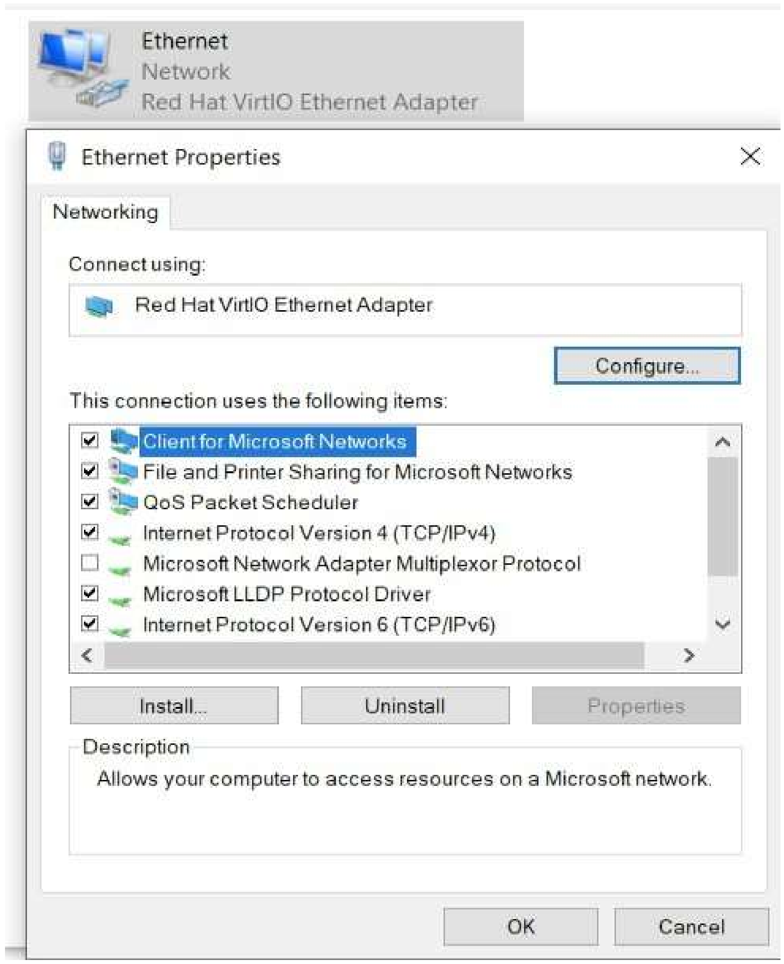
ii. Click on "Network and Internet" Option.



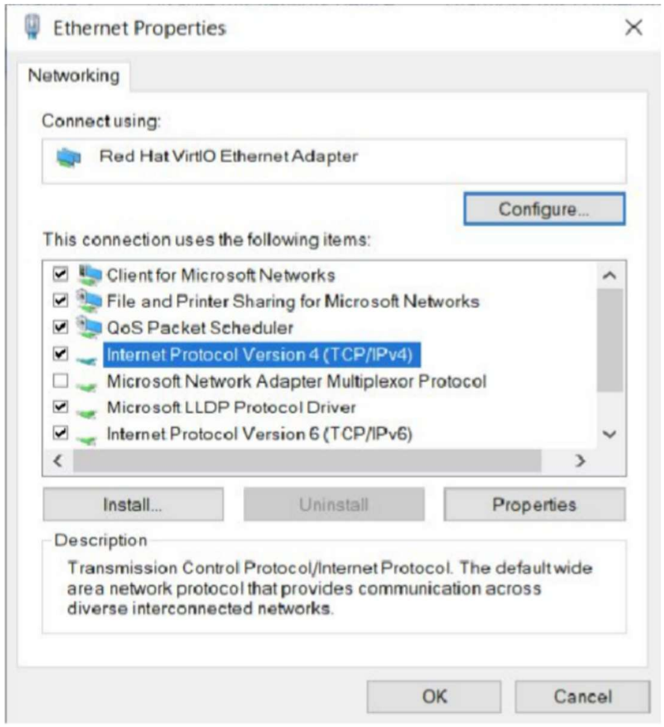
iii. From the window presented, click on "Change Adapter Settings" (upper left side).

3. Select Adapter and Properties:
   i. Various types of connections would be listed (e.g., Wi-Fi, Ethernet).

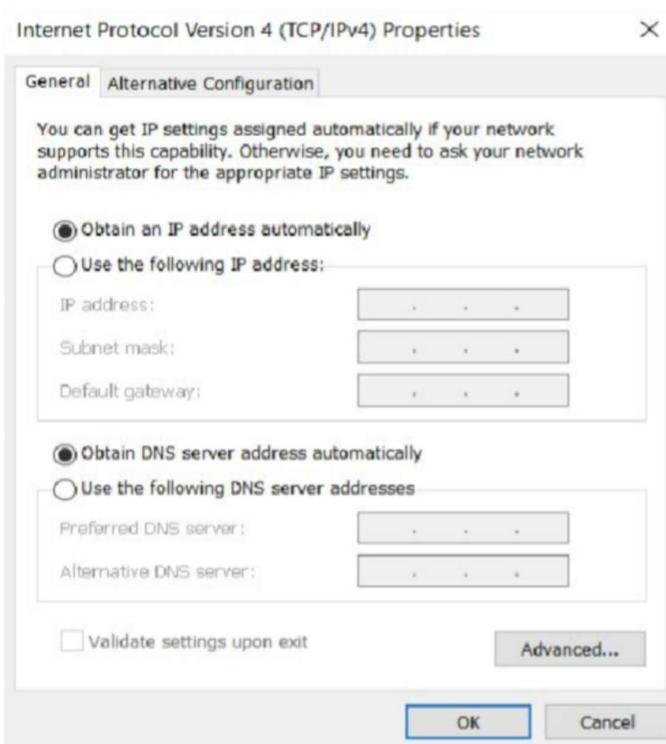
ii. Right-click on the desired network adapter and select "Properties."

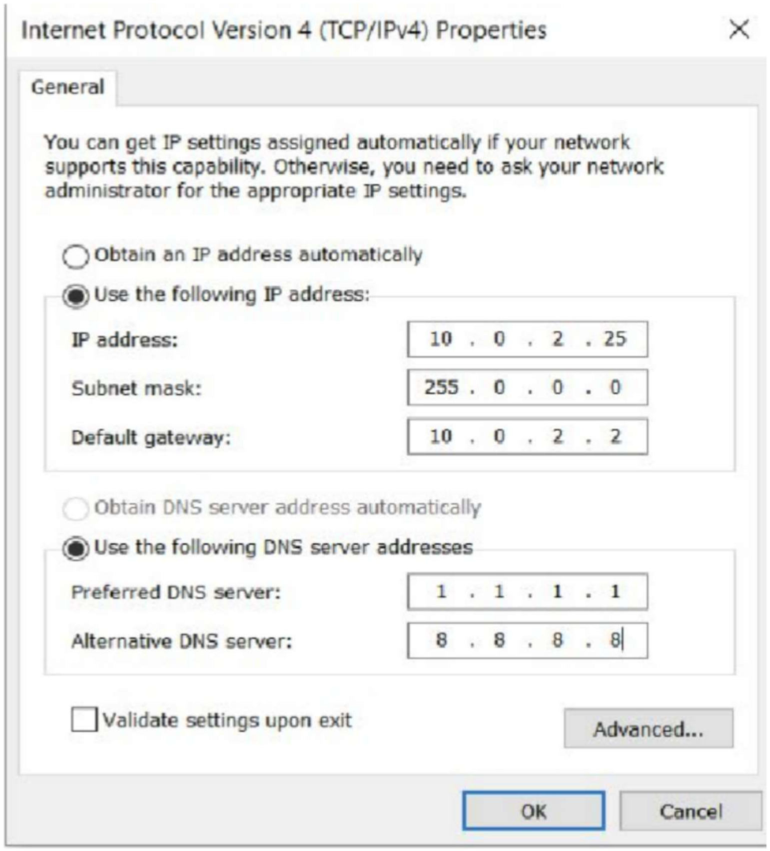


4. Configure IPv4 Properties:
   i. Select "Internet Protocol Version 4 (TCP/IPv4)" and then click "Properties."



ii. Now change the IP address and DNS address as your liking by clicking the radio buttons (e.g., "Use the following IP address" and "Use the following DNS server addresses").





5. Verify Changes:
    i. Finally, test the IP using the ipconfig command.

```
C:\Users\Quickemu>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Site-local IPv6 Address . . . . . : fec0::8112:a59c:20ae:8105%1
   Link-local IPv6 Address . . . . . : fe80::c04a:f179:36c3:c6d2%6
   IPv4 Address. . . . . . . . . . . : 10.0.2.25
   Subnet Mask . . . . . . . . . . . : 255.0.0.0
   Default Gateway . . . . . . . . . : fe80::2%6
                                       10.0.2.2
```

ii. Also, test the connection using ping.

```
C:\Users\Quickemu>ping fifa.com

Pinging fifa.com [23.44.4.177] with 32 bytes of data:
Reply from 23.44.4.177: bytes=32 time=77ms TTL=255
Reply from 23.44.4.177: bytes=32 time=77ms TTL=255
Reply from 23.44.4.177: bytes=32 time=73ms TTL=255
Reply from 23.44.4.177: bytes=32 time=73ms TTL=255

Ping statistics for 23.44.4.177:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 73ms, Maximum = 77ms, Average = 75ms
```

Static IP configuration using command prompt:

1. First, check the current IP using the command *ipconfig /all*

```
C:\Windows\system32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-0ITLP6E
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 52-54-00-22-52-A9
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::ca29:483d:cff0:7c4f%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.122.224(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 July 2025 14:15:44
   Lease Expires . . . . . . . . . . : 12 July 2025 15:15:44
   Default Gateway . . . . . . . . . : 192.168.122.1
   DHCP Server . . . . . . . . . . . : 192.168.122.1
   DHCPv6 IAID . . . . . . . . . . . : 106058752
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-04-10-5F-52-54-00-22-52-A9
   DNS Servers . . . . . . . . . . . : 192.168.122.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

2. Check the connectivity using *ping* command.

```
C:\Windows\system32>ping office.com

Pinging office.com [13.107.6.156] with 32 bytes of data:
Reply from 13.107.6.156: bytes=32 time=42ms TTL=113
Reply from 13.107.6.156: bytes=32 time=47ms TTL=113
Reply from 13.107.6.156: bytes=32 time=46ms TTL=113
Reply from 13.107.6.156: bytes=32 time=46ms TTL=113

Ping statistics for 13.107.6.156:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 42ms, Maximum = 47ms, Average = 45ms
```

3. Get Network Interface Name

```
C:\Windows\system32>netsh interface ipv4 show interfaces

Idx     Met         MTU          State                Name
---  ----------  ----------  ------------  ---------------------------
  1          75  4294967295  connected     Loopback Pseudo-Interface 1
  6          25        1500  connected     Ethernet
```

4. Use the following command to setup static IP and DNS address

```
C:\Windows\system32>netsh interface ipv4 set address name="Ethernet" static 192.168.122.200 255.255.255.0 192.168.122.1

C:\Windows\system32>netsh interface ipv4 set dns name="Ethernet" static 8.8.8.8
```

```
C:\Windows\system32>netsh interface ipv4 add dns name="Ethernet" 1.1.1.1 index=2
```

5. Check the changed IP address.

```
C:\Windows\system32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-0ITLP6E
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 52-54-00-22-52-A9
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::ca29:483d:cff0:7c4f%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.122.200(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.122.1
   DHCPv6 IAID . . . . . . . . . . . : 106058752
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-04-10-5F-52-54-00-22-52-A9
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       1.1.1.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

6. Again, check the connectivity.

```
C:\Windows\system32>ping example.com

Pinging example.com [23.192.228.84] with 32 bytes of data:
Reply from 23.192.228.84: bytes=32 time=260ms TTL=47
Reply from 23.192.228.84: bytes=32 time=257ms TTL=47
Reply from 23.192.228.84: bytes=32 time=261ms TTL=47
Reply from 23.192.228.84: bytes=32 time=259ms TTL=47

Ping statistics for 23.192.228.84:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 257ms, Maximum = 261ms, Average = 259ms
```

7. Revert back to the IP address set by the OS using the following commands.

```
C:\Windows\system32>netsh interface ipv4 set address name="Ethernet" source=dhcp

C:\Windows\system32>netsh interface ipv4 set dns name="Ethernet" source=dhcp
```

8. Lastly,check the IP configuration using the command *ipconfig /all*.

```
C:\Windows\system32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-0ITLP6E
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 52-54-00-22-52-A9
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::ca29:483d:cff0:7c4f%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.122.224(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 July 2025 14:36:48
   Lease Expires . . . . . . . . . . : 12 July 2025 15:36:47
   Default Gateway . . . . . . . . . : 192.168.122.1
   DHCP Server . . . . . . . . . . . : 192.168.122.1
   DHCPv6 IAID . . . . . . . . . . . : 106058752
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-04-10-5F-52-54-00-22-52-A9
   DNS Servers . . . . . . . . . . . : 192.168.122.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Conclusion:

We got an understanding of static IP address configuration on Windows machines. We explored the fundamental concepts of IP addresses (including static and dynamic types), subnet masks, default gateways, and DNS servers. We successfully configured a static IP address using both the Graphical User Interface (GUI) and the Command Prompt methods in Windows. Through initial and final ipconfig and ping commands, we verified that the IP address, subnet mask, default gateway, and DNS settings were correctly applied and that network connectivity was maintained after the changes. Finally, we successfully reverted the network settings back to automatic (DHCP) configuration.